

Food Delivery Time Prediction – ML vs DL Comparative Study

This project presents a comparative analysis of multiple Machine Learning (ML) and Deep Learning (DL) models for predicting food delivery time using different feature sets, feature engineering, and hyperparameter tuning techniques.

Project Files Overview

ML_Model_with_feature_Eng.ipynb

Description: Machine Learning model with feature engineering applied

Columns Used:

- Delivery_person_Age
- Delivery_person_Ratings
- Type_of_order
- Type_of_vehicle
- Time_taken(min)
- Distance
- City
- Region
- Order_Time

Feature Engineering Applied:

- Categorical encoding (City, Region, Vehicle, Order Type)
- Order time extraction
- Distance scaling
- Feature transformation

Result (Test Set):

Model	R ²	RMSE (min)
Extra Trees	0.3422	7.28
Random Forest	0.3405	7.29

Model	R ²	RMSE (min)
CatBoost	0.3407	7.29
LightGBM	0.3312	7.34
Voting Regressor	0.3101	7.45

2 ML_Model_without_FE.ipynb

Description: Machine Learning model without heavy feature engineering

Columns Used:

- Delivery_person_Age
- Delivery_person_Ratings
- Restaurant_latitude
- Restaurant_longitude
- Delivery_location_latitude
- Delivery_location_longitude
- Type_of_order
- Type_of_vehicle
- Time_taken(min)
- Distance

Result (Test Set):

Model	R ²	RMSE
★ Gradient Boosting	0.406	7.23
Voting Regressor	0.403	7.25
CatBoost	0.400	7.26
Extra Trees	0.400	7.26
Random Forest	0.398	7.27
LightGBM	0.396	7.29

Model	R ²	RMSE
XGBoost	0.390	7.33
Decision Tree	0.357	7.52
Linear / Lasso	0.277	7.97

3 ML_Model_(3 input para).ipynb

Description: ML model using only 3 input features

Columns Used:

- Delivery_person_Age
- Delivery_person_Ratings
- Distance

Best Result:

Model	R ²	RMSE
Gradient Boosting	0.3790	7.39

4 DL_model.ipynb

Description: Deep Learning model using full feature set

Result:

Model	R ²	RMSE
LSTM	0.3864	7.34
Simple ANN	0.3846	7.36
Residual NN	0.3846	7.36

5 DL_model(3 input para).ipynb

Description: Deep Learning model using 3 input features

Result:

Model	R ²	RMSE
LSTM	0.3636	7.48
Simple ANN	0.3338	7.65
Residual NN	0.3338	7.65

6 FULL_EDA_Report

Complete exploratory data analysis including:

- Feature distribution
 - Correlation heatmaps
 - Feature importance
 - Trend visualization
 - Outlier detection
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7 ML_model_tune (Hyperparameter Tuning)

Description: ML models trained with tuned hyperparameters for performance optimization.

HistGradientBoostingRegressor (Tuned)

- R² Score: 0.2912
 - MAE: 6.23
 - RMSE: 7.88
 - Accuracy: 0.29
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Random Forest Regressor (Tuned)

- R² Score: 0.2759
- MAE: 6.34
- RMSE: 7.97

XGBoost Regressor (Tuned)

- R^2 Score: 0.1303
 - MAE: 6.83
 - RMSE: 8.73
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Overall Performance Comparison

Approach	Best R^2	Best RMSE
ML without Feature Engineering	★ 0.406	★ 7.23
DL Full Features	0.386	7.34
ML (3 Features)	0.379	7.39
ML with Feature Engineering	0.342	7.28
DL (3 Features)	0.364	7.48
ML Tuned Models	0.291	7.88

Final Best Performing Model

Gradient Boosting Regressor (Without Feature Engineering)

- $R^2 = 0.406$
 - RMSE = 7.23 minutes
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Final Conclusion

This project compared:

- Machine Learning models
- Deep Learning models
- Feature engineering impact
- Reduced feature models
- Hyperparameter tuning

🔥 Key Findings:

1. Traditional ML models outperformed Deep Learning models.
 2. Gradient Boosting achieved the highest R^2 score.
 3. Heavy feature engineering did not improve performance significantly.
 4. Hyperparameter tuning did not outperform the base Gradient Boosting model.
 5. Deep Learning models required higher complexity but did not yield superior results.
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📌 Final Statement

For tabular food delivery datasets, ensemble tree-based Machine Learning models (especially Gradient Boosting) perform better than Deep Learning approaches in terms of accuracy, efficiency, and simplicity.